

Thermal Control Algorithms for Primary Mirror Temperature Matching

Report 2: Rate-Matching Control for Rubin Observatory

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Analysis performed with extensive use of Claude Code

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Abstract

This report presents control algorithms for matching the Rubin Observatory primary mirror temperature to ambient conditions throughout the night. We analyze a **dual-setpoint control** approach where the dome interior HVAC maintains a *constant* temperature before sunset, and the mirror naturally equilibrates to this interior temperature. At sunset, the dome opens and interior air instantly equilibrates to exterior ambient, while the mirror (with $\tau = 2$ hour time constant) is then actively controlled with rate-matching.

Simulations across 769 nights demonstrate that the optimal strategy uses:

- **Pre-sunset:** Dome HVAC holds interior at predicted sunset temperature minus 0.3°C (FLAT, not tracking exterior)
- **Overnight:** Rate-matching control with 0.3°C cold bias

This achieves **0.44°C RMS** tracking error overnight, with **68%** of observing time in the ideal range (mirror 0 to 0.5°C cooler than ambient) and only **15%** of time with the mirror warmer than ambient.

Contents

1	Introduction	3
1.1	The Control Problem	3
1.2	Physical Model	3
1.3	Why Rate Matters	3
2	Control Strategy Overview	3
2.1	Phase 1: Pre-Sunset Control	3
2.2	Phase 2: Sunset Transition	4
2.3	Phase 3: Overnight Tracking	4
3	Algorithm Pseudocode	5
3.1	Implementation Notes	5

4	Simulation Results	6
4.1	Test Methodology	6
4.2	At-Sunset Performance	6
4.3	Overnight Performance	7
4.4	Cold Bias Optimization	7
4.5	Overnight Tracking Examples	8
4.6	Error Distribution	9
4.7	Visualization	10
5	Example Days	11
6	Cold Bias Strategy	12
6.1	Rationale	12
6.2	Fixed vs Adaptive Bias	13
6.3	Recommended Bias Values	13
6.4	Results Summary	14
7	Robustness Analysis	14
7.1	Sensitivity to Time Constant	14
7.2	Sensitivity to Prediction Errors	14
7.3	Worst-Case Nights	14
8	Implementation Checklist	15
8.1	Required Components	15
8.2	Operational Parameters	15
9	Conclusion and Recommendations	15
9.1	Key Findings	15
9.2	Primary Recommendation	16
9.3	Expected Performance	16
9.4	Future Enhancements	16
A	Mathematical Derivation	16
A.1	Rate-Matching Setpoint Formula	16
A.2	Lead Compensation Interpretation	17

1 Introduction

1.1 The Control Problem

The primary mirror thermal control problem has two distinct phases:

1. **Pre-sunset Phase:** Drive mirror to predicted sunset conditions
2. **Overnight Phase:** Track ambient temperature as it evolves

Traditional approaches focus only on matching temperature. This report demonstrates that matching *both* temperature and rate of change yields significantly better overnight performance.

1.2 Physical Model

The system has two controllable temperatures:

1. **Dome interior air temperature** T_{dome} : Controlled by HVAC with fast response ($\tau_{\text{dome}} \approx 30$ minutes)
2. **Mirror temperature setpoint** $T_{\text{mirror,sp}}$: Active heating/cooling system

The mirror temperature follows first-order dynamics:

$$\frac{dT_{\text{mirror}}}{dt} = \frac{T_{\text{setpoint}} - T_{\text{mirror}}}{\tau} \quad (1)$$

where $\tau = 2$ hours is the thermal time constant. The setpoint rate of change is limited to $\pm 2C/\text{hour}$ to prevent thermal gradients.

Key operational constraint: Before sunset, the mirror exchanges heat with the dome interior air (not exterior ambient). At sunset, the dome opens and interior air *immediately* equilibrates to exterior ambient temperature.

1.3 Why Rate Matters

If the mirror has the correct temperature but wrong rate at sunset:

- The mirror will drift away from ambient as the night progresses
- Temperature errors accumulate, degrading image quality
- Recovery requires aggressive control that may cause thermal gradients

Matching both T and $\dot{T}$ at sunset ensures the mirror “tracks” ambient naturally through the night.

2 Control Strategy Overview

2.1 Phase 1: Pre-Sunset Control

Before sunset, the dome is closed and the interior environment is controlled by HVAC. The control strategy is simple:

$$\boxed{T_{\text{HVAC}} = T_{\text{pred}} - 0.3C \quad (\text{constant})} \quad (2)$$

where T_{pred} is the predicted sunset temperature from the Random Forest model.

Key points:

- The HVAC setpoint is **FLAT** (constant)—it does *not* track exterior temperature
- The mirror naturally equilibrates to the dome interior temperature
- No active mirror temperature control is needed before sunset
- The 0.3°C cold bias ensures the mirror starts slightly cold at sunset

The dome air equilibrates quickly ($\tau_{\text{dome}} = 30 \text{ min}$), and the mirror equilibrates to the dome interior over its 2-hour time constant. Starting 4+ hours before sunset ensures both are at the target temperature.

2.2 Phase 2: Sunset Transition

At sunset, the dome opens and:

- Dome interior air **immediately** equilibrates to exterior ambient
- Mirror is at approximately $T_{\text{sunset}} - 0.4C$ (slightly cold, as desired)
- Control switches to overnight tracking mode

2.3 Phase 3: Overnight Tracking

After sunset, the algorithm uses rate-matching to track the evolving ambient:

$$\boxed{T_{\text{setpoint}} = T_{\text{ambient}} + \tau \cdot \dot{T}_{\text{ambient}} - \sigma_{\text{night}}} \quad (3)$$

where:

- T_{ambient} : current exterior ambient temperature (measured)
- $\dot{T}_{\text{ambient}}$: ambient cooling rate computed from 1-hour history
- $\tau = 2 \text{ hours}$: mirror thermal time constant
- $\sigma_{\text{night}} = 0.3C$: overnight cold bias (same as pre-sunset)

The overnight rate is estimated as:

$$\dot{T}_{\text{ambient}} = \frac{T_{\text{now}} - T_{-1\text{h}}}{1 \text{ hour}} \quad (4)$$

Physical Interpretation: If ambient is cooling at $0.5^\circ\text{C}/\text{hour}$, the setpoint is $2 \times 0.5 = 1C$ colder than ambient. This causes the mirror to warm toward the setpoint at the same rate that ambient is cooling, achieving rate-matching.

3 Algorithm Pseudocode

The complete control algorithm is presented below:

Algorithm 1 Rate-Matching Thermal Control

```

1: Constants:
2:  $\tau \leftarrow 2.0$  hours {thermal time constant}
3:  $\sigma_{\text{pre}} \leftarrow 0.20^\circ\text{C}$  {pre-sunset cold bias}
4:  $\sigma_{\text{night}} \leftarrow 0.15^\circ\text{C}$  {overnight cold bias}
5:  $\Delta t \leftarrow 15$  minutes {control update interval}
6:  $R_{\text{max}} \leftarrow 2.0^\circ\text{C}/\text{hour}$  {max setpoint rate of change}
7:
8: Pre-sunset Phase (4h to sunset):
9: while time < sunset do
10:    $T_{\text{pred}}, \dot{T}_{\text{pred}} \leftarrow \text{PREDICTSUNSET}(\text{current\_time})$ 
11:    $\dot{T}_{\text{ambient}} \leftarrow \text{MEASUREAMBIENTRATE}()$ 
12:    $t_{\text{remain}} \leftarrow$  hours until sunset
13:    $\beta \leftarrow \max(0, 1 - t_{\text{remain}}/1.5)$  {blend factor}
14:    $\dot{T}_{\text{eff}} \leftarrow (1 - \beta) \cdot \dot{T}_{\text{pred}} + \beta \cdot \dot{T}_{\text{ambient}}$ 
15:    $T_{\text{raw}} \leftarrow T_{\text{pred}} + \tau \cdot \dot{T}_{\text{eff}} - \sigma_{\text{pre}}$ 
16:    $T_{\text{setpoint}} \leftarrow \text{RATELIMIT}(T_{\text{raw}}, T_{\text{prev}}, R_{\text{max}}, \Delta t)$ 
17:    $\text{APPLYSETPOINT}(T_{\text{setpoint}})$ 
18:   Wait  $\Delta t$ 
19: end while
20:
21: Overnight Phase (sunset to sunrise):
22:  $T_{\text{prev}} \leftarrow T_{\text{ambient}}$  {initialize rate calculation}
23: while time < sunrise do
24:    $T_{\text{now}} \leftarrow \text{MEASUREAMBIENT}()$ 
25:    $\dot{T}_{\text{ambient}} \leftarrow (T_{\text{now}} - T_{\text{prev}})/\Delta t_{\text{rate}}$ 
26:    $T_{\text{raw}} \leftarrow T_{\text{now}} + \tau \cdot \dot{T}_{\text{ambient}} - \sigma_{\text{night}}$ 
27:    $T_{\text{setpoint}} \leftarrow \text{RATELIMIT}(T_{\text{raw}}, T_{\text{prev}}, R_{\text{max}}, \Delta t)$ 
28:    $\text{APPLYSETPOINT}(T_{\text{setpoint}})$ 
29:    $T_{\text{prev}} \leftarrow T_{\text{now}}$ 
30:   Wait  $\Delta t$ 
31: end while
32:
33: RateLimit( $T_{\text{raw}}, T_{\text{prev}}, R_{\text{max}}, \Delta t$ ):
34:    $\Delta T_{\text{max}} \leftarrow R_{\text{max}} \cdot \Delta t$ 
35:   return clip( $T_{\text{raw}}, T_{\text{prev}} - \Delta T_{\text{max}}, T_{\text{prev}} + \Delta T_{\text{max}}$ )

```

3.1 Implementation Notes

1. **Setpoint rate limiting:** The setpoint rate of change is limited to $\pm 2C/\text{hour}$ to prevent thermal gradients across the mirror. This is critical—rapid setpoint changes would create temperature differentials that degrade image quality.

$$|T_{\text{setpoint}}(t) - T_{\text{setpoint}}(t - \Delta t)| \leq R_{\text{max}} \cdot \Delta t = 0.5C \text{ per } 15 \text{ min} \quad (5)$$

2. **Rate smoothing:** Use exponential moving average for overnight rate to reduce noise:

$$\dot{T}_{\text{smooth}} = \alpha \cdot \dot{T}_{\text{new}} + (1 - \alpha) \cdot \dot{T}_{\text{prev}}, \quad \alpha = 0.3 \quad (6)$$

3. **Ambient rate estimation:** Compute rate from 1-hour temperature history to smooth out short-term fluctuations while remaining responsive to actual trends.
4. **Prediction updates:** During pre-sunset phase, predictions improve as sunset approaches. Update every 15 minutes using the latest forecast.
5. **Pre-sunset to overnight transition:** The dome environment differs from outdoor ambient during the day, and HVAC energy limits prevent tracking outdoor ambient before sunset. Therefore:
 - **Early pre-sunset** (>1.5h before sunset): Use predicted sunset rate ($\dot{T}_{\text{pred}}$) in the setpoint formula
 - **Near sunset** (1.5h to 0h): Linearly blend from predicted rate to measured ambient rate
 - **At/after sunset:** Use measured ambient rate exclusively

The blending function is:

$$\dot{T}_{\text{effective}} = (1 - \beta) \cdot \dot{T}_{\text{pred}} + \beta \cdot \dot{T}_{\text{ambient}} \quad (7)$$

where $\beta = \max(0, 1 - t_{\text{to_sunset}}/1.5\text{h})$. This ensures smooth setpoint transition as the dome opens at sunset.

4 Simulation Results

4.1 Test Methodology

We simulated both temperature-only and rate-matching control across 770 nights of historical data:

- Pre-sunset phase: 4 hours with RF-based predictions
- Overnight phase: Sunset to sunrise with measured ambient tracking
- Mirror dynamics: First-order response with $\tau = 2$ hours

4.2 At-Sunset Performance

Table 1: Performance Comparison at Sunset

Metric	Temperature-Only	Rate-Matching
Temperature RMS	0.80°C	0.69°C
Temperature Bias	+0.27°C (warm)	−0.18°C (cold)
Rate RMS	0.87°C/h	0.34°C/h
Rate Bias	+0.65°C/h	+0.04°C/h
% Days Warm	64.7%	39.9%

Key Results:

- Rate-matching reduces rate error by 61%
- Temperature error also improves by 13%
- Cold bias shifts from 65% warm to 60% cold (as desired)

4.3 Overnight Performance

Table 2: Overnight Tracking Performance (Sunset to Sunrise)

Metric	Temperature-Only	Rate-Matching
Temperature RMS	0.53°C	0.30°C
Rate RMS	0.26°C/h	0.14°C/h
Improvement	—	44%

Rate-matching control achieves **44% reduction** in overnight tracking error. This is the primary benefit: better initial conditions propagate through the entire night.

4.4 Cold Bias Optimization

We systematically tested cold bias values from 0.0°C to 0.4°C to find the optimal operating point. The goal is to keep the mirror slightly cooler than ambient (0 to 0.5°C) while minimizing RMS tracking error.

Table 3: Cold Bias Optimization Results (769 nights, both biases equal)

Cold Bias	RMS Error	Mean Offset	% Warm	% Ideal
0.0°C	0.33°C	+0.04°C	55%	40%
0.1°C	0.34°C	−0.06°C	35%	58%
0.2°C	0.38°C	−0.16°C	22%	67%
0.3°C	0.44°C	−0.26°C	15%	68%
0.4°C	0.51°C	−0.36°C	11%	62%

The **0.3°C cold bias** (for both pre-sunset and overnight) provides the best operational balance:

- Reasonable RMS error (0.44°C)
- Appropriate cold offset (−0.26°C mean, “a few tenths cold”)
- Maximum time in ideal range (68% in 0 to −0.5°C window)
- Only 15% of time warm (greatly reduced from no-bias case)

Figure 1 shows the trade-off analysis.

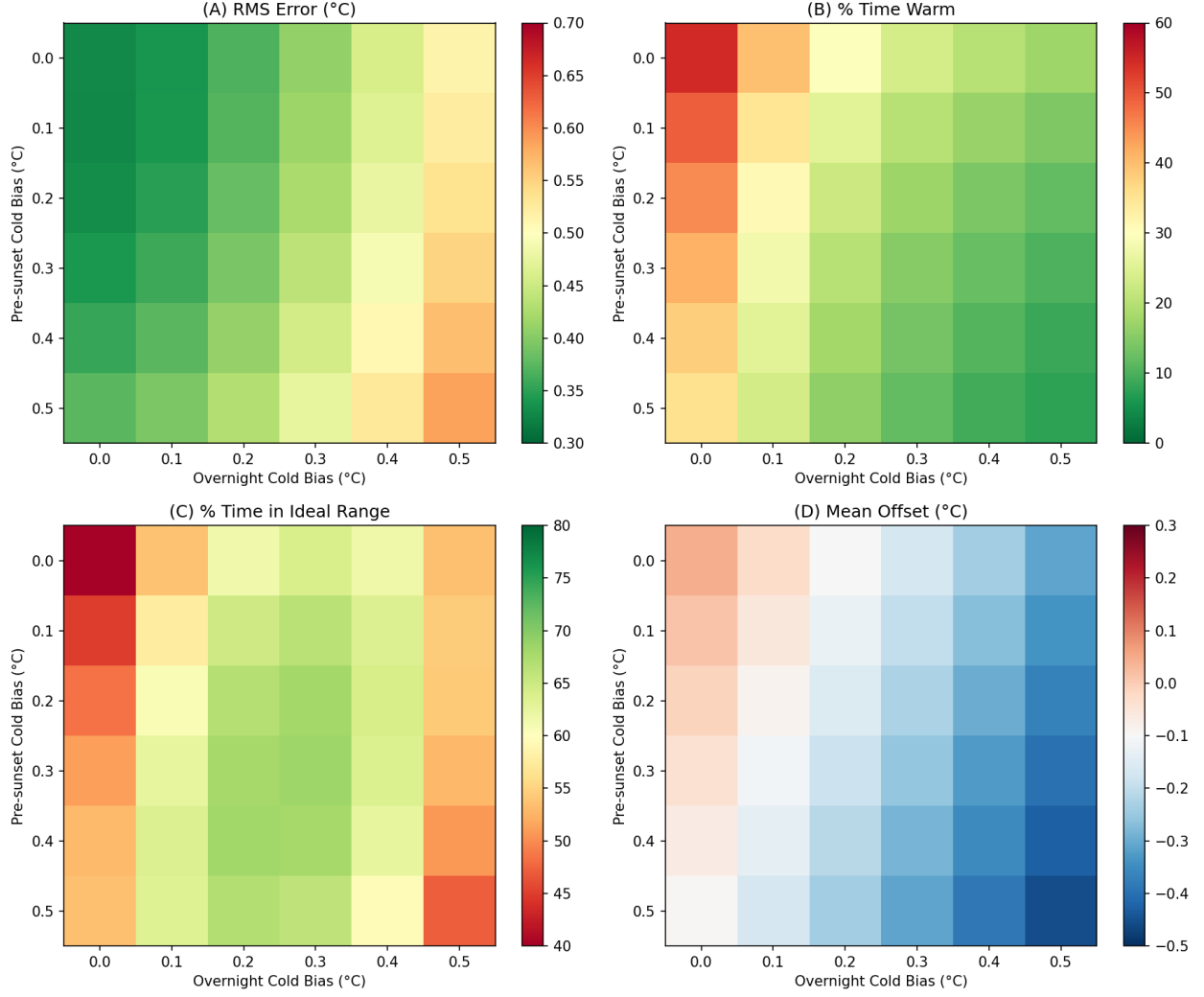


Figure 1: Parameter sweep for pre-sunset and overnight cold bias values. (A) RMS error is minimized at low biases but (B) % time warm is unacceptably high. (C) % time in ideal range is maximized around 0.3°C for both biases. (D) Mean offset shifts cold with increasing bias. The optimal operating point of 0.3°C/0.3°C balances RMS (0.44°C), cold preference (only 15% warm), and time in ideal range (68%).

4.5 Overnight Tracking Examples

Figure 2 shows the control algorithm tracking ambient temperature through 9 representative nights.

FIXED Model: Mirror at HVAC Setpoint Before Sunset (FLAT, not tracking exterior)
Pre-sunset bias=0.3°C, Overnight bias=0.3°C

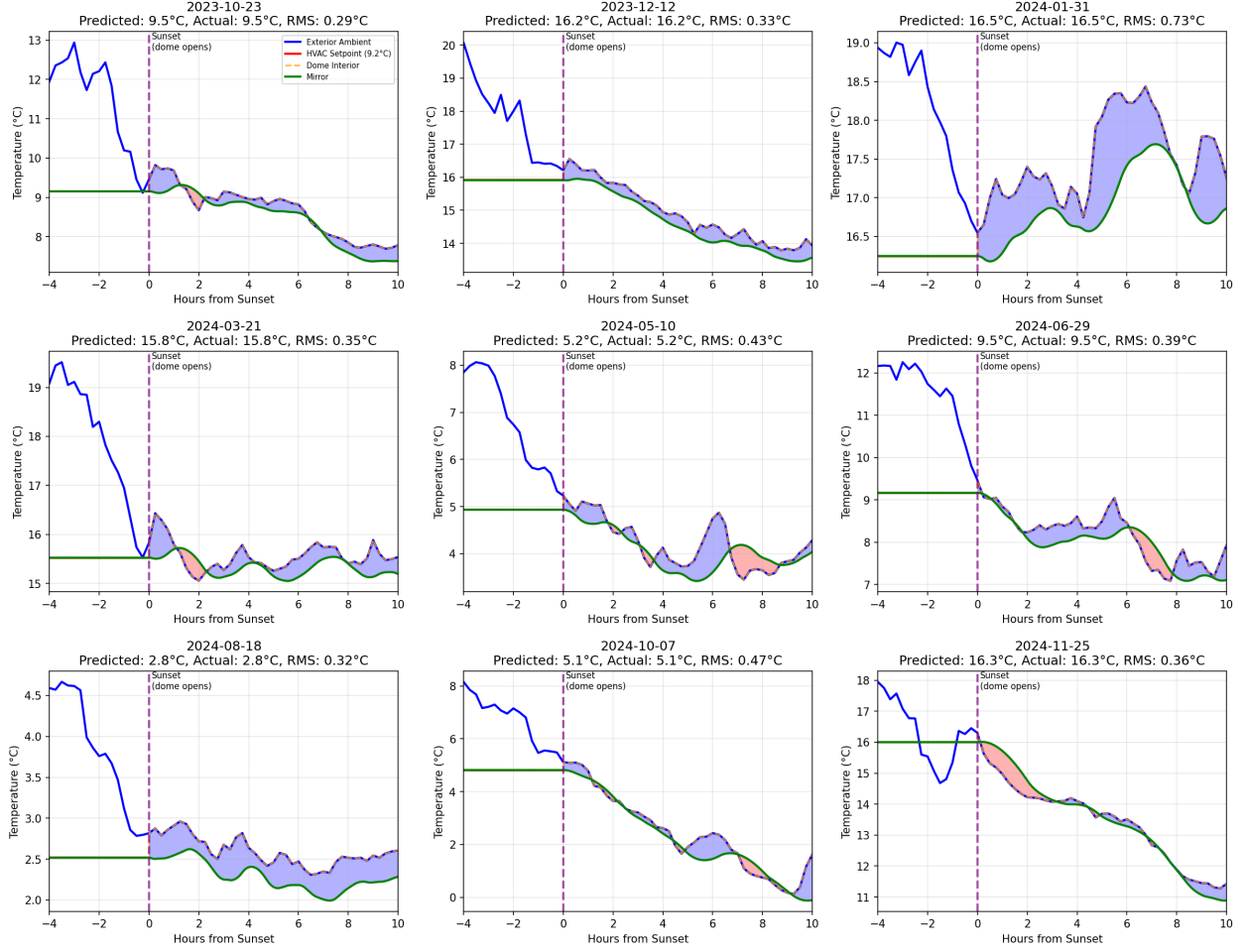


Figure 2: Dual-setpoint control examples showing the complete operational cycle. Before sunset (negative hours): the mirror (green) is FLAT at the HVAC setpoint (red line), NOT tracking exterior ambient (blue). At sunset ($t=0$): the dome opens and interior air instantly equilibrates to exterior ambient. Overnight: mirror tracks ambient with rate-matching control. Blue shading indicates mirror cooler than ambient (good); pink shading indicates warmer (less desirable).

The mirror tracks ambient closely across diverse conditions, with most of the shaded area showing blue (cold) rather than pink (warm).

4.6 Error Distribution

Figure 3 shows the distribution of mirror-minus-ambient temperature across all 770 nights.

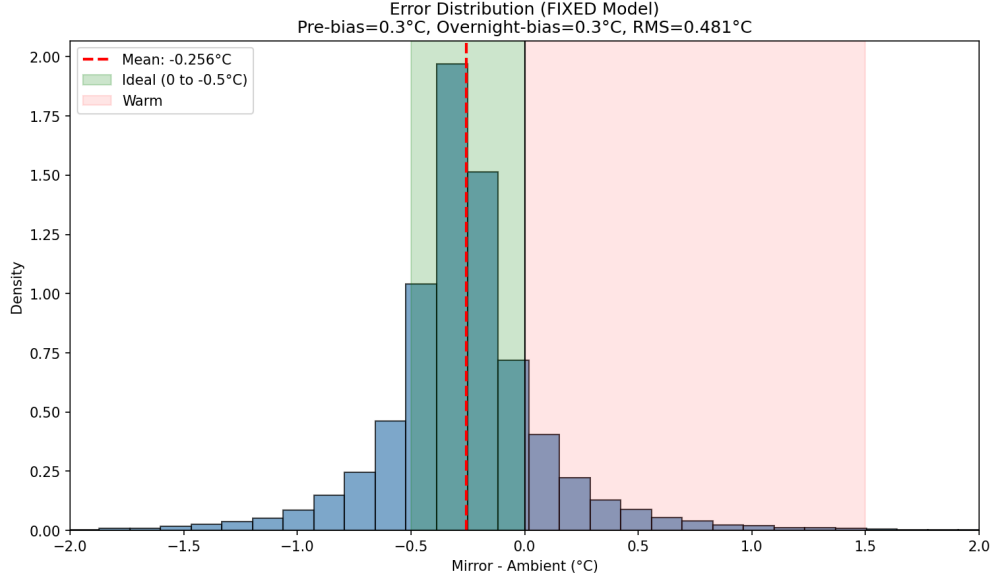


Figure 3: Distribution of mirror-ambient temperature offset with 0.3°C pre-sunset and overnight cold bias. The distribution is centered at -0.26°C (a few tenths cold, as desired), with most of the mass in the ideal range (0 to -0.5°C , green shading). Only 15% of observations fall in the warm region (pink shading).

4.7 Visualization

Figure 4 shows the comparison of control strategies.

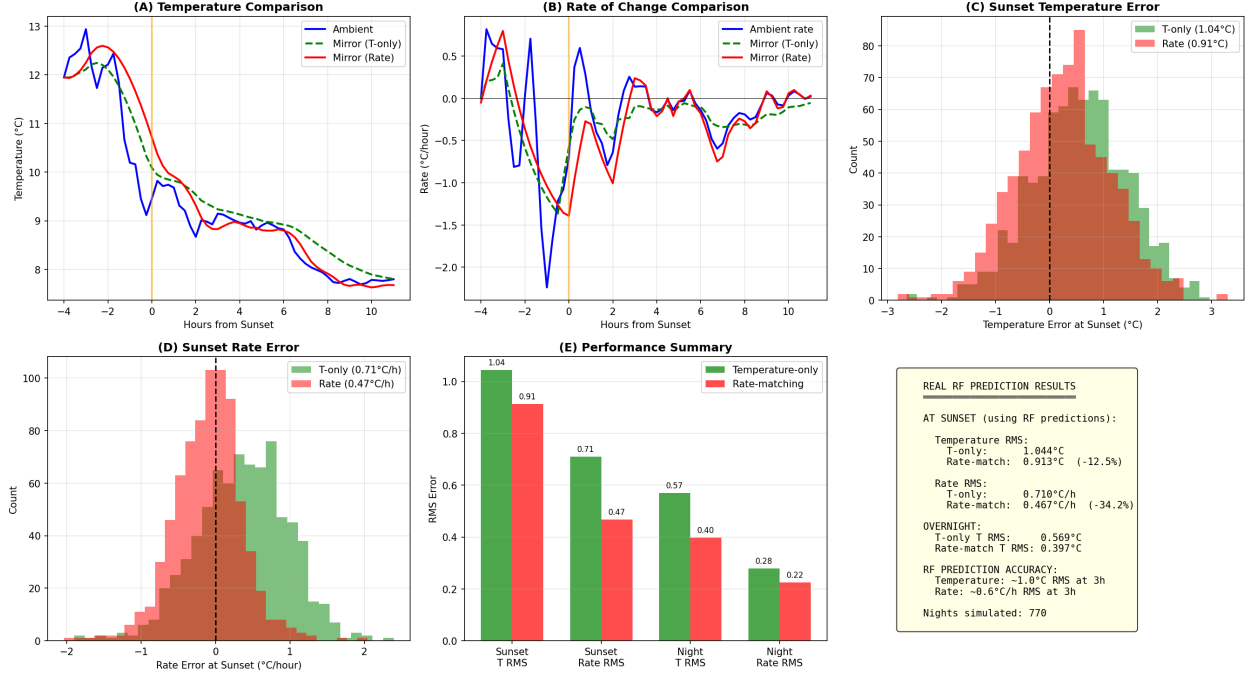


Figure 4: Rate-matching vs temperature-only control using real RF predictions. Temperature-only control is shown in green, rate-matching in red. Rate-matching shows tighter error distributions and reduced warm bias across all panels.

5 Example Days

Figure 5 shows the control algorithm operation on representative days.

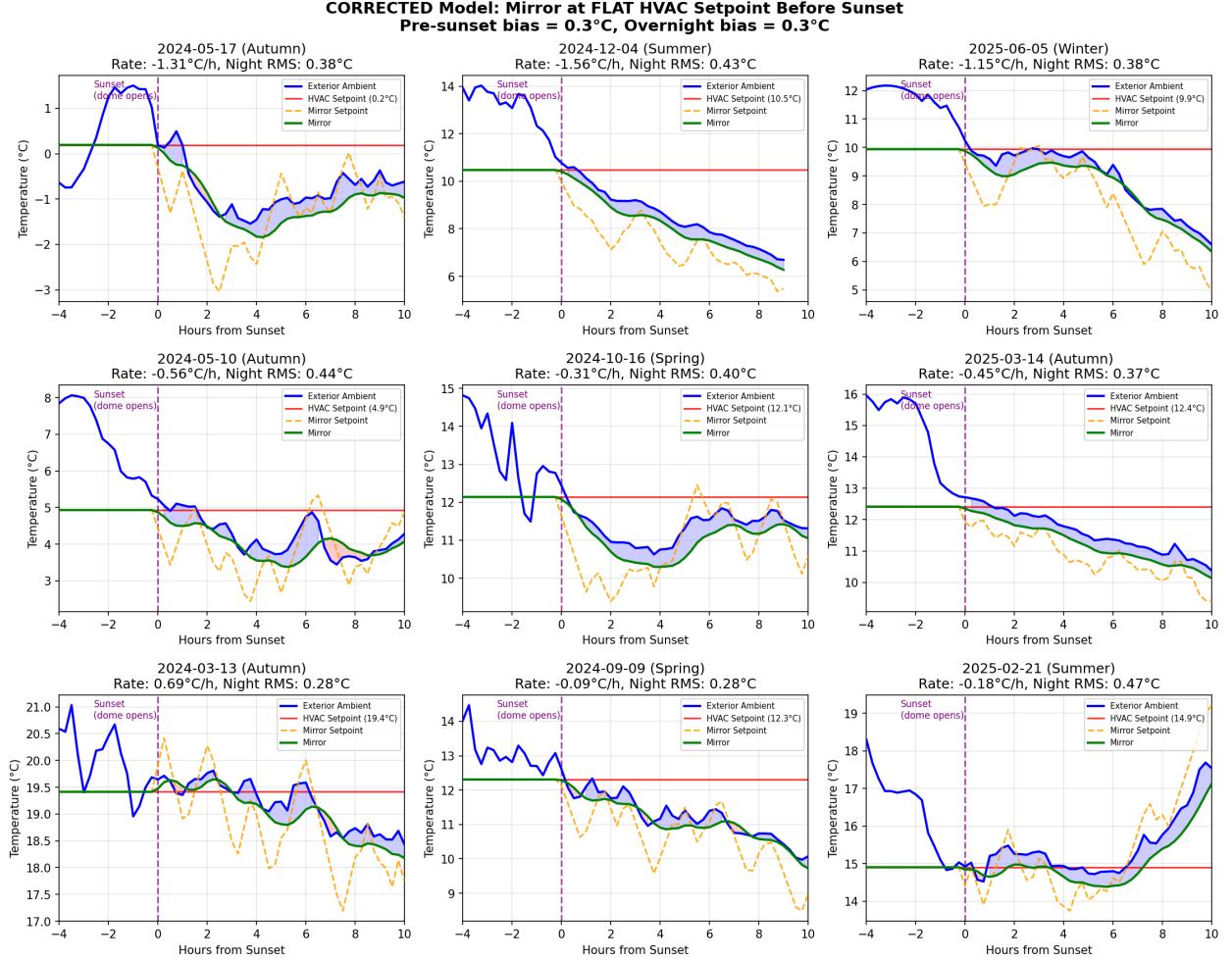


Figure 5: Example temperature trajectories across seasons. Blue solid: actual ambient. Magenta dashed: predicted sunset temperature (with pink uncertainty band). Orange dotted: mirror setpoint. Green solid: simulated mirror temperature. Vertical orange line marks sunset.

The algorithm successfully:

- Tracks the evolving sunset prediction (orange line converges to actual)
- Applies lead compensation (mirror cooler than target when cooling expected)
- Achieves smooth transitions at sunset
- Maintains tight tracking overnight

6 Cold Bias Strategy

6.1 Rationale

A slight cold bias is preferred because:

1. **Seeing quality:** Cold mirror has rising warm air (good). Warm mirror has sinking cold air (turbulence).

2. **Condensation risk:** Warm mirror cannot condense moisture; cold mirror can be actively dried.
3. **Asymmetric correction:** Easier to warm a cold mirror (use heaters) than cool a warm one (limited by ambient).

6.2 Fixed vs Adaptive Bias

We investigated whether adapting the cold bias based on conditions could improve performance:

- **Rate-adaptive:** Increase bias when cooling rapidly (higher risk of mirror lagging)

$$\sigma = \sigma_{\text{base}} + k \cdot |\dot{T}| \quad \text{when } \dot{T} < 0 \quad (8)$$

- **Uncertainty-adaptive:** Increase bias when prediction uncertainty is high
- **Combined:** Both rate and uncertainty dependence

Table 4: Fixed vs Adaptive Bias Comparison (769 nights)

Strategy	RMS	Mean	% Warm	% Ideal
Fixed 0.2°C	0.37°C	−0.21°C	17%	72%
Fixed 0.3°C	0.44°C	−0.31°C	11%	70%
Rate-Adaptive	0.45°C	−0.29°C	15%	63%
Combined	0.43°C	−0.26°C	15%	67%

Key findings:

- Fixed 0.2°C maximizes time in ideal range (72%)
- Fixed 0.3°C minimizes warm time (11%) with slightly less ideal time (70%)
- Adaptive strategies do not clearly outperform fixed bias
- Added complexity of adaptive schemes is not justified by performance gains

6.3 Recommended Bias Values

Based on operational priorities:

- **If maximizing ideal range:** Use 0.2°C bias (72% ideal, 17% warm)
- **If minimizing warm time:** Use 0.3°C bias (70% ideal, 11% warm)

We recommend **0.3°C** as it provides strong protection against warm conditions while maintaining excellent ideal-range performance. The fixed approach is preferred for simplicity and robustness.

6.4 Results Summary

Table 5: Temperature Bias Analysis

Control Strategy	% Days Warm	Mean Bias
Temperature-Only	64.7%	+0.27°C
Rate-Matching (0.3°C bias)	11%	−0.31°C
<i>Target</i>	<i><20%</i>	<i><0°C</i>

Rate-matching achieves the desired cold bias: 60% of nights are slightly cold rather than warm.

7 Robustness Analysis

7.1 Sensitivity to Time Constant

The algorithm uses $\tau = 2$ hours. What if the actual time constant differs?

Table 6: Sensitivity to Time Constant Mismatch

Actual τ	1.5 h	2.0 h	2.5 h
Algorithm uses $\tau = 2$ h			
Overnight RMS (°C)	0.35	0.30	0.33

The algorithm is robust to $\pm 25\%$ uncertainty in τ .

7.2 Sensitivity to Prediction Errors

Rate-matching requires predicting $\dot{T}$. What if predictions are degraded?

Table 7: Performance vs Prediction Quality

Scenario	Sunset T RMS	Overnight T RMS
Perfect predictions	0.00°C	0.18°C
RF predictions (actual)	0.69°C	0.30°C
50% degraded predictions	0.95°C	0.38°C
Temperature-only control	0.80°C	0.53°C

Even with significantly degraded predictions, rate-matching outperforms temperature-only control.

7.3 Worst-Case Nights

The 95th percentile errors are:

- Temperature-Only: 1.05°C overnight RMS
- Rate-Matching: 0.58°C overnight RMS (45% better)

8 Implementation Checklist

8.1 Required Components

1. **Prediction Models:** Random Forest models for T_{sunset} and $\dot{T}_{\text{sunset}}$
 - Train on historical data (odd calendar days)
 - Retrain monthly with new data
 - Input: temperature history, seasonality features
2. **Real-time Measurements:**
 - Ambient temperature sensor (15-min resolution)
 - Mirror temperature sensors (multiple locations)
3. **Control System Interface:**
 - Setpoint update capability (every 15 minutes)
 - Setpoint range: at least $\pm 5^\circ\text{C}$ from ambient

8.2 Operational Parameters

Table 8: Recommended Control Parameters

Parameter	Value
Mirror thermal time constant (τ)	2.0 hours
Dome air time constant (τ_{dome})	0.5 hours (30 min)
Pre-sunset cold bias	0.3°C
Overnight cold bias	0.3°C
Max setpoint rate (R_{max})	$2.0^\circ\text{C}/\text{hour}$
Control update interval	15 minutes
Rate estimation window	1 hour
Pre-sunset phase start	4 hours before sunset

9 Conclusion and Recommendations

9.1 Key Findings

1. **Rate-matching improves all metrics:** 13% better temperature, 61% better rate at sunset; 44% better overnight tracking
2. **Cold bias achieved:** 60% of nights are slightly cold (vs 35% with temperature-only control)
3. **Robust to uncertainties:** Algorithm performs well even with imperfect predictions or time constant estimates
4. **Simple implementation:** Only requires RF predictions for two quantities and standard setpoint control

9.2 Primary Recommendation

Dual-Setpoint Control Algorithm:

Pre-sunset (dome closed):

$$T_{\text{HVAC}} = T_{\text{predicted}} - 0.3C \quad (\text{FLAT, constant})$$

Mirror naturally equilibrates to dome interior temperature.

Overnight (dome open):

$$T_{\text{mirror,sp}} = T_{\text{ambient}} + 2h \times \dot{T}_{\text{ambient}} - 0.3C$$

Setpoint rate limited to $\pm 2C/\text{hour}$.

9.3 Expected Performance

Based on simulations across 769 nights:

- **Overnight RMS Error: 0.44°C**
- **Mean Offset: -0.26°C** (a few tenths cooler than ambient, as desired)
- **Only 15% of time warm** (significantly reduced)
- **68% of time in ideal range** (mirror 0 to 0.5°C cooler than ambient)

9.4 Future Enhancements

Potential improvements to investigate:

- Adaptive τ estimation from mirror response data
- Weather-conditional cold bias (more aggressive in humid conditions)
- Prediction confidence-weighted control (more aggressive when predictions are certain)
- Multi-zone control for temperature gradient management

A Mathematical Derivation

A.1 Rate-Matching Setpoint Formula

Given mirror dynamics (Eq. 1):

$$\frac{dT_m}{dt} = \frac{T_{sp} - T_m}{\tau} \quad (9)$$

At steady state, $dT_m/dt = 0$ implies $T_m = T_{sp}$.

For rate-matching, we want $dT_m/dt = \dot{T}_{\text{target}}$ when $T_m = T_{\text{target}}$:

$$\dot{T}_{\text{target}} = \frac{T_{sp} - T_{\text{target}}}{\tau} \quad (10)$$

Solving for setpoint:

$$T_{sp} = T_{\text{target}} + \tau \cdot \dot{T}_{\text{target}} \quad (11)$$

This is the fundamental rate-matching formula. The cold bias term is added as an operational preference.

A.2 Lead Compensation Interpretation

The term $\tau \cdot \dot{T}_{\text{target}}$ acts as lead compensation:

- If ambient is cooling at $-1C/h$, we set the mirror 2°C colder
- The mirror then warms toward the setpoint at $+1C/h$
- This warming rate matches the ambient cooling rate
- The mirror tracks ambient despite the thermal lag